- HDOC : DAY 14 -

Program for Digit Analysis

1. Problem Statement

Write a menu-driven C program using switch-case to take a positive integer as input and perform one of the following operations based on the user's choice:

1. Check whether all its digits are unique.
2. Find the frequency of each digit from '0' to '9'.

2. Algorithm

Main Program Algorithm:

1. Start.
2. Declare variables: num, choice.
3. Display the menu options to the user:
    a. Press 1 to check if digits are unique.
    b. Press 2 to find digit frequency.
4. Input choice from the user.
5. Input a positive integer num.
6. Use switch(choice):
    a. **Case 1:** Execute logic to check if all digits are unique.
    b. **Case 2:** Execute logic to count frequency of digits 0 to 9.
    c. **Default:** Display "Invalid choice".
7. Stop.

Algorithm for Choice 1 (Check Unique Digits):

1. Take a copy of num, say temp.
2. Initialize an array count[10] with all elements set to 0.
3. Set isUnique = 1 (assume digits are unique).
4. While temp > 0:
    a. Extract the last digit: rem = temp % 10.

b. Increment count[rem].
c. If count[rem] > 1, set isUnique = 0 and break the loop.
d. Reduce number: temp = temp / 10.

5. If isUnique == 1, print "Unique".
6. Else, print "Not Unique".

Algorithm for Choice 2 (Digit Frequency):

1. Take a copy of num, say temp.
2. Initialize an array freq[10] with all elements set to 0.
3. While temp > 0:
   a. Extract last digit: rem = temp % 10.
   b. Increment freq[rem]++.
   c. Reduce number: temp = temp / 10.
4. Print the frequency of each digit from 0 to 9 in the format freq_0, freq_1, ..., freq_9.

3. Test Case Table

Test case	Input choice	Input Number	Expected output	Actual op	status
1	1	4819	Output: Unique	4819	pass
2	1	7552	Output: Not Unique	7552	pass
3	2	50558	freq_0 = 1, freq_1 = 0, freq_2 = 0, freq_3 = 0, freq_4 = 0, freq_5 = 3, freq_6 =	freq_0 = 1, freq_1 = 0, freq_2 = 0, freq_3 = 0, freq_4 = 0, freq_5 = 3, freq_6 =	pass

			0, freq_7 = 0, freq_8 = 1, freq_9 = 0	0, freq_7 = 0, freq_8 = 1, freq_9 = 0	

4. C Program Code

```
#include <stdio.h>

int main() {
    int choice;
    int num, temp, rem;
    int freq[10] = {0};
    int isUnique = 1;
    int i;

    printf("--- MENU ---\n");
    printf("1. Check whether all digits are unique\n");
    printf("2. Find frequency of each digit (0-9)\n");
    printf("Enter your choice (1 or 2): ");
    scanf("%d", &choice);

    printf("Enter a positive integer: ");
    scanf("%d", &num);

    switch(choice) {
        case 1:
            temp = num;
            while(temp > 0) {
                rem = temp % 10;
                freq[rem]++;
                if(freq[rem] > 1) {
                    isUnique = 0;
                    break;
                }
                temp = temp / 10;
            }

            if(isUnique == 1) {
                printf("Output: Unique\n");
            } else {
                printf("Output: Not Unique\n");
            }
            break;
```

```c
        case 2:
            temp = num;
            while(temp > 0) {
                rem = temp % 10;
                freq[rem]++;
                temp = temp / 10;
            }

            for(i = 0; i < 10; i++) {
                printf("freq_%d = %d", i, freq[i]);
                if(i < 9) {
                    printf(", ");
                }
            }
            printf("\n");
            break;

        default:
            printf("Invalid choice!\n");
    }

    return 0;
}
```

EXECUTION AND OUTPUT :

```
manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % nano 14q11.c
manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % gcc 14q11.c -o 14q11

manalithakar@Unknown_8e:80:9d:9c:62:11 ~ %
manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % ./14q11
--- MENU ---
1. Check whether all digits are unique
2. Find frequency of each digit (0-9)
Enter your choice (1 or 2): 1
Enter a positive integer: 4819
Output: Unique
manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % ./14q11
--- MENU ---
1. Check whether all digits are unique
2. Find frequency of each digit (0-9)
Enter your choice (1 or 2): 1
Enter a positive integer: 7552
Output: Not Unique
manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % ./14q11
--- MENU ---
1. Check whether all digits are unique
2. Find frequency of each digit (0-9)
Enter your choice (1 or 2): 2
Enter a positive integer: 50558
freq_0 = 1, freq_1 = 0, freq_2 = 0, freq_3 = 0, freq_4 = 0, freq_5 = 3, freq_6 = 0, freq_7 = 0, freq_8 = 1, freq_9 = 0
manalithakar@Unknown_8e:80:9d:9c:62:11 ~ %
```